1) $\int \sin^2 x \cos^6 x, dx$

$\sin^2 x = 1 - \cos^2 x$

$\int (1-\cos^2 x)\cos^6 x\, dx$

$\int (\cos^6 x - \cos^8 x)\, dx$

$\cos^2 x = \frac{1+\cos 2x}{2}$

$\cos^6 x = (\cos^2 x)^3 = \left(\frac{1+\cos 2x}{2}\right)^3$

$\cos x = (\cos^2 x)^4 = \left(\frac{1+\cos 2x}{2}\right)^4$

$= \frac{1}{128}(5x + 2\sin 2x - \sin 4x -$

$\frac{2}{3}\sin 6x - \frac{1}{8}\sin 8x) + C$

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3) $\int \sin^3 x \cos^3 x, dx$

$\int \sin^3 x \cos^2 x (\cos x\, dx)$

Let $u = \sin x$ Let $du = \cos x\, dx$

$\cos^2 x = 1 - \sin^2 x = 1 - u^2$

$= \int u^3(1-u^2)\, du$

$\int (u^3 - u^5)\, du$

$= \frac{u^4}{4} - \frac{u^6}{6} + C$

$= \frac{\sin^4 x}{4} - \frac{\sin^6 x}{6} + C$

+C

$\int \tan^4 x \, dx$

$\tan^2 x = \sec^2 x - 1$

$\tan^4 x = (\sec^2 x - 1)^2$

$\sec^4 x - 2\sec^2 x + 1$

$\int \tan^4 x dx = \int \sec^4 x dx - 2\int \sec^2 x dx +$

$\int 1 dx$

$= \int \sec^2 x dx = \tan x +$

For $\int \sec^2 x \cdot \sec^2 x dx$

Let $u = \tan x$, $du = \sec^2 x dx$

So:

$= \int (1 + \tan^2 x) \sec^2 x dx =$

$\int (1 + u^2) du$

$= u + \frac{u^3}{3} = \tan x + \frac{\tan^3 x}{3}$

$\left(\tan x + \frac{\tan^3 x}{3}\right) - 2\tan x + x + C$

$= x - \tan x + \frac{\tan^3 x}{3} + C$

$$\int \frac{\sin^2(\sqrt{x})}{\sqrt{x}}\,dx$$

Let $u = \sqrt{x} \Rightarrow x = u^2, \quad dx = 2u\,du$

$$\int \frac{\sin^2 u}{u}(2u\,du)$$

$$= 2\int \sin^2 u\,du$$

$$\sin^2 u = \frac{1-\cos 2u}{2}$$

so;

$$2\cdot\int \frac{1-\cos 2u}{2}\,du$$

$$= \int (1-\cos 2u)\,du$$

$$= u - \frac{\sin 2u}{2} + C$$

$$= \sqrt{x} - \frac{1}{2}\sin(2\sqrt{x}) + C$$

$\int \cot^4 x \csc^2 x \, dx$

$\frac{d}{dx}(\cot x) = -\csc^2 x$

Let $u = \cot x \Rightarrow du = -\csc^2 x\, dx$

So:

$= -\int u^4 \, du$

$= \frac{-u^5}{5} + C$

$= -\frac{\cot^5 x}{5} + C$